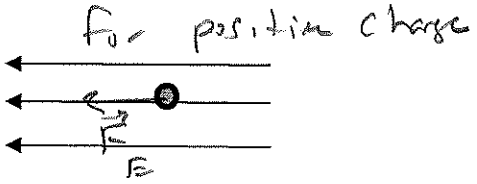
Practice Problems
Calculus-Based Physics II

Name: Key **Date:** _____

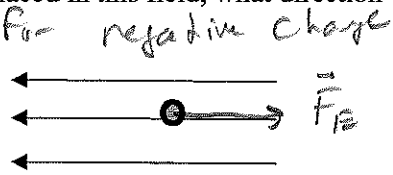
Chapter 36 Part A Electric Fields

1) An electric field is pointing to the left, an object with a net positive charge is placed in this field, what direction does the object experience a force?



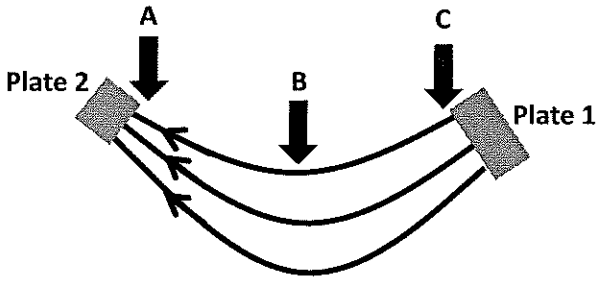
(A) To the left.
B) To the right.
C) It does not experience a force.
D) Not enough information is given to know what direction.

2) An electric field is pointing to the left, an object with a net negative charge is placed in this field, what direction does the object experience a force?



A) To the left.
(B) To the right.
C) It does not experience a force.
D) Not enough information is given to know what direction.

3) An electric field is in a space between plate 1 and plate 2 in the direction from right to left, which section of the electric field indicated by the letters (A, B or C) has the largest field strength?



(A) Section A is largest
B) Section B is largest
C) Section C is largest
D) They are all the same field strength

4) Using the diagram above, what must the charge on Plate 2 be in order to create this electric field?

A) Positive
(B) Negative
C) Neutral
D) No way to determine the charge on Plate 2

$\vec{E}$ - field → From positive to negative charge

5) A proton and an electron are placed in an unform electric field. Which undergoes the greater acceleration?

A) proton
(B) electron
C) Both accelerate equally.
D) Neither accelerates.

$$\Sigma \vec{F} = m\vec{a} = q\vec{E}$$

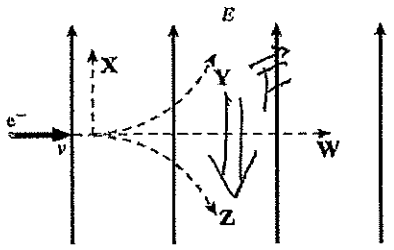
$$\Sigma F_x = ma_x = qE_x$$

$$a_x = \frac{\Sigma F_x}{m} = \frac{qE_x}{m}$$

negative charge

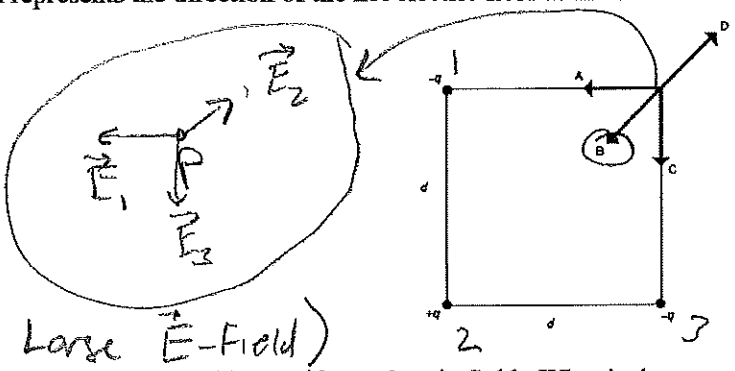
6) An electron is initially moving to the right when it enters a uniform electric field directed upwards, as shown in the figure. Which trajectory (X, Y, Z, or W) will the electron follow in the field?

A) trajectory **W**
B) trajectory **X**
C) trajectory **Y**
D) trajectory **Z**



7) Three equal point charges of varying signs are placed on three of the corners of a square of side *d* as shown in the figure below. Which of the arrows shown represents the direction of the net electric field at the vacant corner of the square?

A) A
B) B
C) C
D) D
E) none of the above



(q_1 + q_3 will produce Large $\vec{E}$-Field)

8) A force of 6.0 N acts on a charge of 3.0 μC when it is placed in a uniform electric field. What is the magnitude of this electric field?

$\Sigma \vec{F} = \vec{F}_E = q\vec{E}$

$\Sigma F_x = F_{Ex} = qE_x$

$E_x = \frac{F_x}{q} = 2 \times 10^6$ N/C

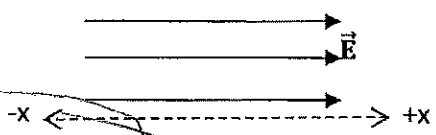
9) A proton is placed in an electric field of intensity 700 N/C. What is the magnitude and direction of the acceleration of this proton due to this field? mass of proton = 1.67×10^{-27} kg, $q_p = 1.6 \times 10^{-19}$ C

$\Sigma \vec{F} = m\vec{a} = \vec{F}_E$

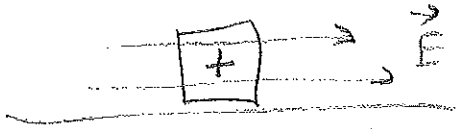
$F_x = ma_x = qE_x$

$a_x = \frac{qE_x}{m}$

$a_x = 6.7 \times 10^{10}$ m/s²



10) A small object with a 5.0-μC charge is accelerating horizontally on a friction-free surface at 0.0050 m/s² due only to an electric field. If the object has a mass of 2.0 g, what is the magnitude of the electric field?



$\Sigma \vec{F} = \vec{F}_E = m\vec{a}$

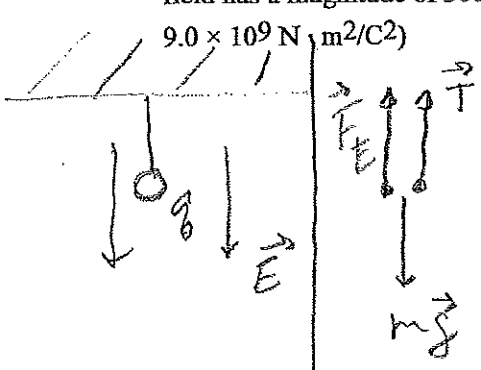
$\Sigma F_x = F_x = ma_x$

$F_x = qE_x$

$qE_x = ma_x$

$E_x = \frac{ma_x}{q}$ = 2 N/C

11) A small 0.050-kg insulating sphere carries a negative charge of 60 μC and is hanging by a vertical silk thread from a fixed point in the ceiling. An external uniform vertical electric field is now applied. If the applied electric field has a magnitude of 3000 N/C and is directed downward, what is the tension in the silk thread? ($k = 1/4\pi\varepsilon_0 = 9.0 \times 10^9$ N, m^2/C^2)



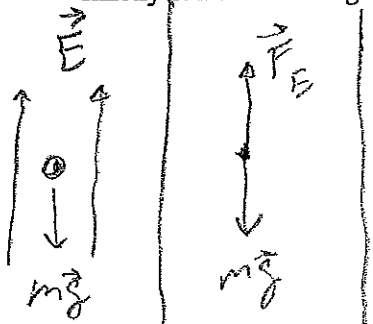
$\Sigma \vec{F} = \vec{F}_E + \vec{T} + m\vec{g} = 0$

$\Sigma F_y = F_E + T - mg = 0$

$T = mg - F_E = mg - qE_y$

$T = 0.49N - 0.18N = 0.31N$

12) A particle with a positive charge of 4.0 μC has a mass of 5.0×10^{-3} kg. What electric field directed upward will exactly balance the weight of the particle and in what direction (up or down)?

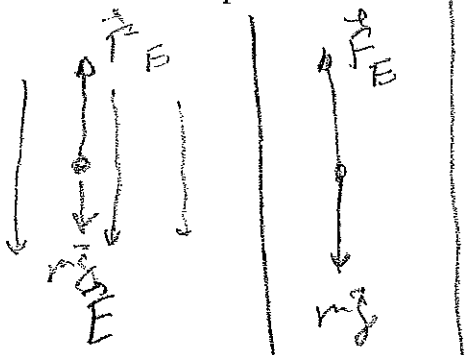


$\Sigma \vec{F} = \vec{F}_E + m\vec{g} = 0$

$\Sigma F_y = F_E - mg = 0 = qE_y - mg$

$E_y = \frac{mg}{q} = 12,250\ N/C$

13) A Styrofoam ball of mass 0.120 g is placed in an electric field of 6000 N/C pointing downward. What charge must be placed on the ball for it to be suspended?



$\Sigma \vec{F} = \vec{F}_E + m\vec{g} = 0$

$\Sigma F_y = F_E - mg = 0 = qE_y - mg = 0$

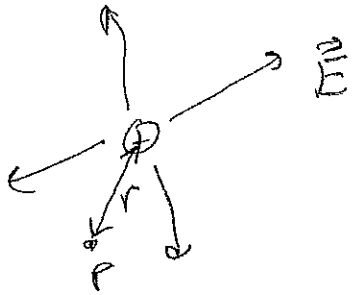
$q = \frac{mg}{E_y} = 1.96 \times 10^{-7}\ C$

14) What are the magnitude and direction of the electric field at a distance of 1.5 m from a positive 50 nC charged sphere?

For point source $\vec{E} = \left(\frac{Kq}{r^2}\right)\hat{r}$

$|\vec{E}| = \frac{Kq}{r^2} = 200\ N/C$

point away from "+" charge



15) A small glass bead has been charged to 1.9 nC. What is the strength of the electric field 2.0 cm from the center of the bead? ($k = 1/4\pi\varepsilon_0 = 8.99 \times 10^9\ \text{N} \cdot \text{m}^2/\text{C}^2$)

$\vec{E} = \left(\frac{kq}{r^2}\right)\hat{i} \Rightarrow |\vec{E}| = \frac{kq}{r^2} = 42{,}750\ \text{N/C}$

16) The electric field at a point 2.8 cm from a small object points toward the object with a strength of 180,000 N/C. What is the object's charge q? ($k = 1/4\pi\varepsilon_0 = 8.99 \times 10^9\ \text{N} \cdot \text{m}^2/\text{C}^2$)

q ⊖ ← $\vec{E}$ P

$|\vec{E}| = \frac{kq}{r^2}$

$q = \frac{|\vec{E}|\, r^2}{k} = 1.57 \times 10^{-8}\ \text{C}$

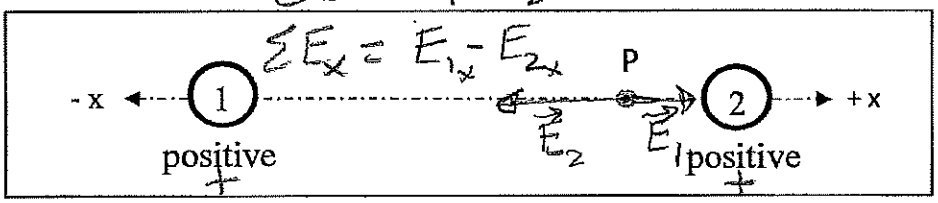
17) Two identically charged objects are shown below, what is the direction of the electric field at point P due to the two charged objects?

$\Sigma\vec{E} = \vec{E}_1 + \vec{E}_2$

A) Positive x direction
B) Negative x direction
C) Zero electric field

$\Sigma E_x = E_{1x} - E_{2x}$

$|\vec{E}_2| > |\vec{E}_1|$

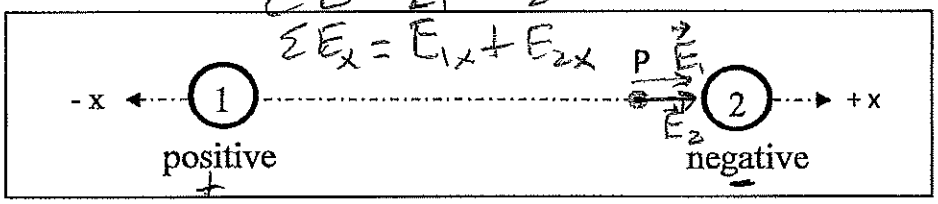


18) Two identically charged objects are shown below, what is the direction of the electric field at point P due to the two charged objects?

$\Sigma\vec{E} = \vec{E}_1 + \vec{E}_2$

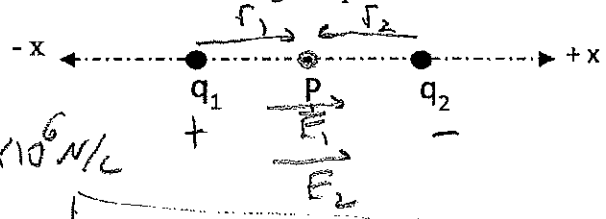
$\Sigma E_x = E_{1x} + E_{2x}$

A) Positive x direction
B) Negative x direction
C) Zero electric field



19) Two charged spheres with a net charge of $q_1 = 20\ \mu\text{C}$ positive and $q_2 = 8\ \mu\text{C}$ negative, are separated by a distance of 20 cm. What is the magnitude of electric field midway between these two charges at point P?

$\Sigma\vec{E} = \vec{E}_1 + \vec{E}_2$



$\Sigma E_x = |\vec{E}_1| + |\vec{E}_2| = 1.8 \times 10^7\ \text{N/C} + 7.2 \times 10^6\ \text{N/C}$

$\Sigma E_x = 2.52 \times 10^7\ \text{N/C}$

$|\vec{E}_1| = \frac{kq_1}{r_1^2} = 1.8 \times 10^7\ \text{N/C}$

$|\vec{E}_2| = \frac{kq_2}{r_2^2} = 7.2 \times 10^6\ \text{N/C}$

20) Two charged objects are placed along the x axis, $q_1 = 2\ \mu C$ positive at the origin and $q_2 = 4\ \mu C$ negative and placed at the 1 m mark, what is the magnitude of the electric field at the 0.2 meter mark indicated by point P



$\Sigma \vec{E} = \vec{E}_1 + \vec{E}_2$

$\Sigma E_x = E_{1x} + E_{2x}$

$\Sigma E_x = 4.5 \times 10^5 N/C + 5.6 \times 10^4 N/C$

$\Sigma E_x = 5.06 \times 10^5 N/C$

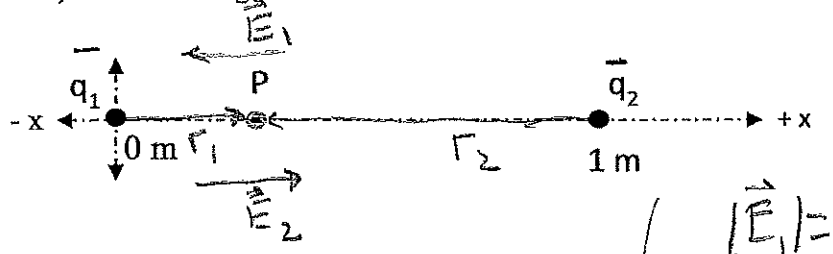
$r_1 = 0.2 m$

$r_2 = 0.8 m$

$|\vec{E}_1| = \frac{kq_1}{r_1^2} = 4.5 \times 10^5 N/C$

$|\vec{E}_2| = \frac{kq_2}{r_2^2} = 5.6 \times 10^4 N/C$

21) Two charged objects are placed along the x axis, $q_1 = 2\ \mu C$ **negative** at the origin and $q_2 = 4\ \mu C$ negative and placed at the 1 m mark, what is the magnitude of the electric field at the 0.2 meter mark indicated by point P



$\Sigma \vec{E} = \vec{E}_1 + \vec{E}_2$

$\Sigma E_x = E_{2x} - E_{1x} = 5.6 \times 10^4 N/C - 4.5 \times 10^5 N/C$

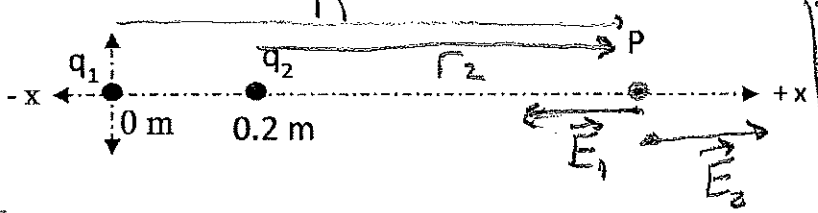
$\Sigma E_x = -3.94 \times 10^5 N/C$

↖ also negative x direction

$|\vec{E}_1| = \frac{kq_1}{r_1^2} = 4.5 \times 10^5 N/C$

$|\vec{E}_2| = \frac{kq_2}{r_2^2} = 5.6 \times 10^4 N/C$

22) Two charged objects are placed along the x axis, $q_1 = 2\ \mu C$ **negative** at the origin and $q_2 = 4\ \mu C$ positive and placed at the 0.2 m mark, what is the magnitude of the electric field at the 1 meter mark indicated by point P



$\Sigma \vec{E} = \vec{E}_1 + \vec{E}_2$

$\Sigma E_x = E_{2x} - E_{1x}$

$\Sigma E_x = 5.63 \times 10^4 N/C - 1.8 \times 10^4 N/C$

$\Sigma E_x = +3.83 \times 10^4 N/C$

↖ + x direction

$r_1 = 1 m$

$r_2 = 0.8 m$

$|\vec{E}_1| = \frac{kq_1}{r_1^2} = 1.8 \times 10^4 N/C$

$|\vec{E}_2| = \frac{kq_2}{r_2^2} = 5.63 \times 10^4 N/C$

23) Two tiny particles having charges of $q_1 = 7\ \mu C$ positive and $q_2 = ?$ are placed along the x-axis. The positive 7 μC particle is at $x = 0$ cm, and q_2 is at $x = +40$ cm. If the net electric field strength is zero at point P located at -3 m what must the charge on q_2 be and what type of charge (positive or negative)?

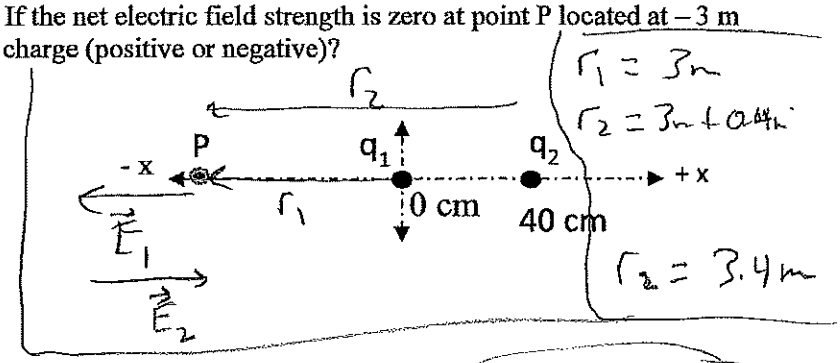
$$\Sigma \vec{E} = \vec{E}_1 + \vec{E}_2 = 0$$

$$\Sigma E_x = (-E_1) + E_2 = 0$$

$$\Sigma E_x = -\frac{Kq_1}{r_1^2} + \frac{Kq_2}{r_2^2} = 0$$

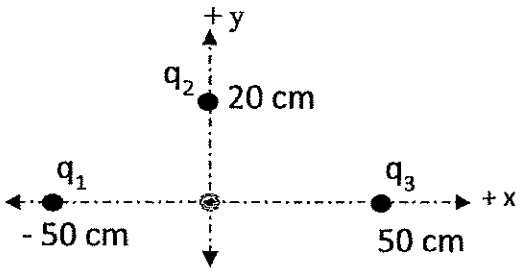
Solve for q_2

$$\frac{Kq_2}{r_2^2} = \frac{Kq_1}{r_1^2}$$



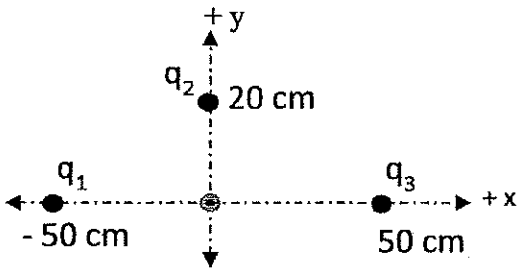
$$q_2 = \frac{r_2^2 q_1}{r_1^2} = 9.0 \times 10^{-6}\ C$$

24) Three charged spheres are placed as shown, determine the magnitude and direction of the electric field at the origin of the coordinate system if $q_1 = 8\ \mu C$ positive, $q_2 = 6\ \mu C$ negative, and $q_3 = 8\ \mu C$ positive.



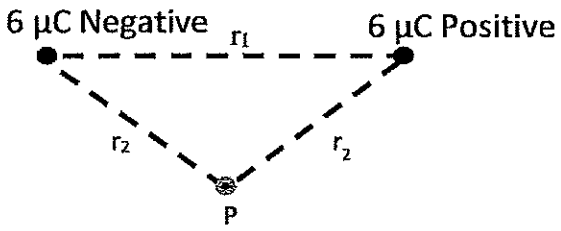
See attached

25) Three charged spheres are placed as shown, determine the magnitude and direction of the electric field at the origin of the coordinate system if $q_1 = 8\ \mu C$ positive, $q_2 = 6\ \mu C$ negative, and $q_3 = 8\ \mu C$ **negative**.



See attached

26) An electric dipole consists of two charged objects, if the net charge on the first object is a negative 6.00 μC and the second has a net charge of positive 6 μC, they are $r_1 = 10$ cm apart, as shown in the figure. Find the magnitude and direction of the electric field this dipole produces at point P, which is $r_2 = 7.00$ cm from each charge. ($k = 9 \times 10^9\ N \cdot m^2/C^2$)

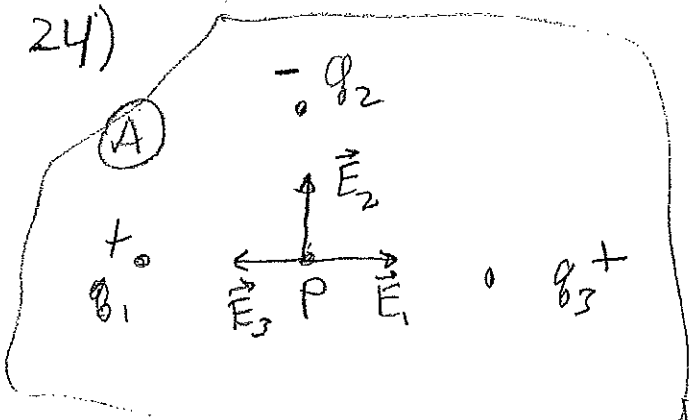


See Attached

Answers

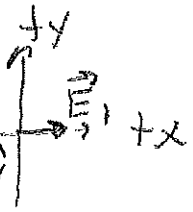
1) A
2) B
3) A
4) B
5) B
6) D
7) B
8) 2×10^6 N/C
9) 6.7×10^{10} m/s^2
10) 2 N/C
11) 0.31 N
12) 12,250 N/C
13) 1.96×10^{-7} C
14) 200 N/C directed away from sphere
15) 42,750 N/C
16) 1.57×10^{-8} C
17) B
18) A
19) 2.52×10^7 N/C
20) 5.06×10^5 N/C
21) 3.94×10^5 N/C in the negative x direction
22) 3.83×10^4 N/C in the positive x direction
23) 9.0×10^{-6} C
24) 1.35×10^6 N/C along positive y axis $\alpha = 90°$ and $\beta = 0°$
25) 1.47×10^6 N/C $\alpha = 67°$ and $\beta = 23°$
26) 1.57×10^7 N/C along negative x axis $\alpha = 180°$ and $\beta = 90°$

24)



$$\Sigma\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3$$

(B) need vector $\vec{E}_1$ in compact form.



$$\vec{E}_1 = (|\vec{E}_1|\cos(\alpha))\hat{x} + (|\vec{E}_1|\cos(\beta))\hat{y}$$

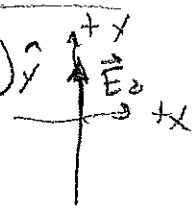
$\alpha = 0°$ $\beta = 90°$

$$|\vec{E}_1| = \frac{kq_1}{r_1^2} = 2.88\times10^5\ N/C$$

$$\vec{E}_1 = (2.88\times10^5\ N/C)\hat{x} + (0\ N/C)\hat{y}$$

$$|\vec{E}_2| = \frac{kq_2}{r_2^2} = 1.35\times10^6\ N/C$$

$$\vec{E}_2 = (|\vec{E}_2|\cos(\alpha))\hat{x} + (|\vec{E}_2|\cos(\beta))\hat{y}$$

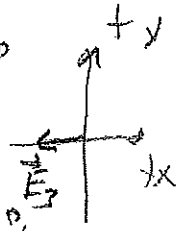


$\alpha = 90°$ $\beta = 0°$

$$\vec{E}_2 = (0\ N/C)\hat{x} + (1.35\times10^6\ N/C)\hat{y}$$

$$|\vec{E}_3| = \frac{kq_3}{r_3^2} = 2.88\times10^5\ N/C$$

$$\vec{E}_3 = (|\vec{E}_3|\cos(\alpha))\hat{x} + (|\vec{E}_3|\cos(\beta))\hat{y}$$



$\alpha = 180°$ $\beta = 90°$

$$\vec{E}_3 = (-2.88\times10^5\ N/C)\hat{x} + (0\ N/C)\hat{y}$$

(C) $\Sigma\vec{E} = (\Sigma E_x)\hat{x} + (\Sigma E_y)\hat{y}$

$$\Sigma E_x = 2.88\times10^5\ N/C + 0\ N/C + -2.88\times10^5\ N/C$$

$$\Sigma E_x = 0\ N/C$$

$$\Sigma E_y = 0\ N/C + 1.35\times10^6\ N/C + 0\ N/C$$

$$\Sigma E_y = 1.35\times10^6\ N/C$$

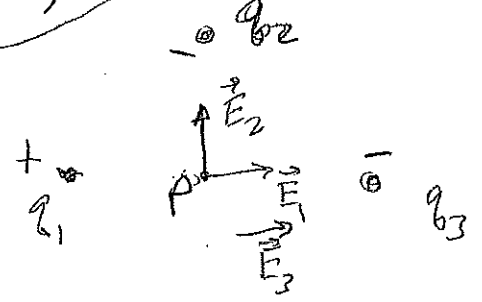
(D) $\Sigma\vec{E} = (\Sigma E_x)\hat{x} + (\Sigma E_y)\hat{y} = (0\ N/C)\hat{x} + (1.35\times10^6\ N/C)\hat{y}$

$$|\Sigma\vec{E}| = \sqrt{E_x^2 + E_y^2} = 1.35\times10^6\ N/C$$

$$\alpha = \cos^{-1}\left(\frac{E_x}{|\Sigma\vec{E}|}\right) = 90°$$

$$\beta = \cos^{-1}\left(\frac{E_y}{|\Sigma\vec{E}|}\right) = 0°$$

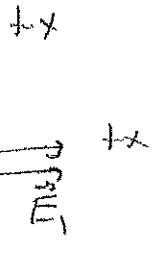
25)



$$\Sigma \vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3$$

$$\vec{E}_1 = (|\vec{E}_1|\cos(\alpha))\hat{x} + (|\vec{E}_1|\cos(\beta))\hat{y}$$

$\alpha = 0°$ $\quad \beta = 90°$



$$\vec{E}_1 = (2.88\times10^5 N/C)\hat{x} + (0 N/C)\hat{y}$$

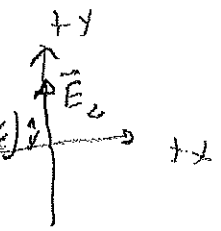
$$|\vec{E}_1| = \frac{kq_1}{r_1^2} = 2.88\times10^5 N/C$$

$$|\vec{E}_2| = \frac{kq_2}{r_2^2} = 1.35\times10^6 N/C$$

$$|\vec{E}_3| = \frac{kq_3}{r_3^2} = 2.88\times10^5 N/C$$

$$\vec{E}_2 = (|\vec{E}_2|\cos(\alpha))\hat{x} + (|\vec{E}_2|\cos(\beta))\hat{y}$$

$\alpha = 90°$ $\quad \beta = 0°$

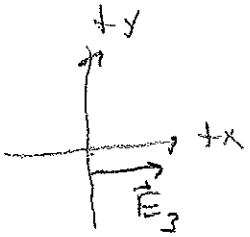


$$\vec{E}_2 = (0 N/C)\hat{x} + (1.35\times10^6 N/C)\hat{y}$$

$$\vec{E}_3 = (|\vec{E}_3|\cos(\alpha))\hat{x} + (|\vec{E}_3|\cos(\beta))\hat{y}$$

$\alpha = 0°$ $\quad \beta = 90°$

$$\vec{E}_3 = (2.88\times10^5 N/C)\hat{x} + (0 N/C)\hat{y}$$



$$\Sigma \vec{E} = (\Sigma E_x)\hat{x} + (\Sigma E_y)\hat{y}$$

$$\Sigma E_x = 2.88\times10^5 N/C + 2.88\times10^5 N/C$$

$$\Sigma E_x = +5.76\times10^5 N/C$$

$$\Sigma E_y = 0 N/C + 1.35\times10^6 N/C + 0 N/C$$

$$\Sigma E_y = 1.35\times10^6 N/C$$

$$|\Sigma \vec{E}| = \sqrt{E_x^2 + E_y^2}$$

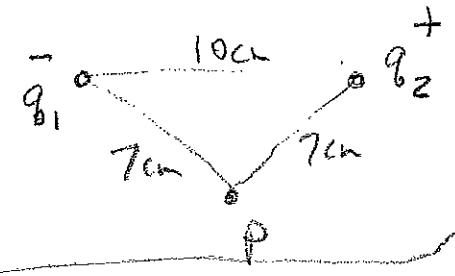
$$|\Sigma \vec{E}| = 1.47\times10^6 N/C$$

$$\alpha = \cos^{-1}\left(\frac{E_x}{|\Sigma \vec{E}|}\right) = 67°$$

$$\beta = \cos^{-1}\left(\frac{E_y}{|\Sigma \vec{E}|}\right) = 23°$$

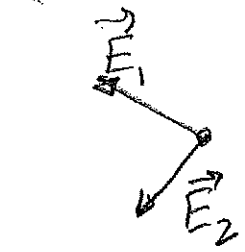
$$\Sigma \vec{E} = (5.76\times10^5 N/C)\hat{x} + (1.35\times10^6 N/C)\hat{y}$$

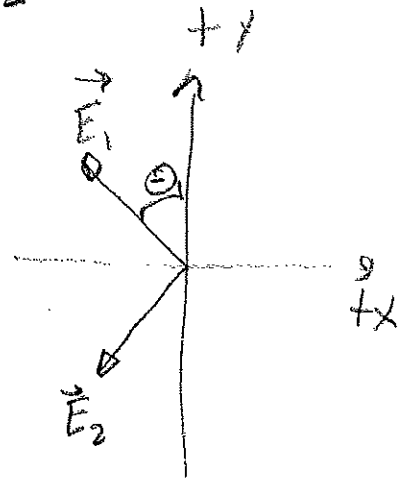
2b)



$\Sigma \vec{E}_p = \vec{E}_1 + \vec{E}_2$

at P





need vector components

$\Sigma \vec{E}_p \cdot \hat{x} = \vec{E}_1 \cdot \hat{x} + \vec{E}_2 \cdot \hat{x}$

$\Sigma E_x = |\vec{E}_1| \cos(\alpha_1) + |\vec{E}_2| \cos(\alpha_2)$

$\alpha_1 = 90° + \theta_1 = 135.6°$

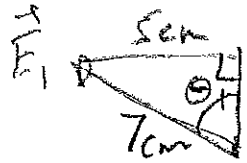
$\alpha_2 = 180° + \theta_2 = 224.4°$

$\Sigma E_x = -7.86 \times 10^6 N/C + (-7.6 \times 10^6 N/C)$

$\Sigma E_x = -1.57 \times 10^7 N/C$

$|\vec{E}_1| = \frac{k q_1}{r_1^2} = 1.1 \times 10^7 N/C$

$|\vec{E}_2| = \frac{k q_2}{r_2^2} = 1.1 \times 10^7 N/C$



$\sin\theta = \frac{5cm}{7cm}$

$\theta_1 = \sin^{-1}\left(\frac{5cm}{7cm}\right)$

$\theta_1 = 45.6°$

$\Sigma \vec{E}_p \cdot \hat{y} = \vec{E}_1 \cdot \hat{y} + \vec{E}_2 \cdot \hat{y} = \Sigma E_y$

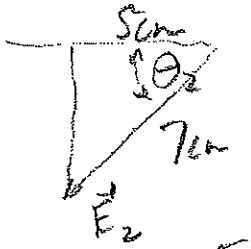
$\beta_1 = \theta_1 = 45.6°$

$\beta_2 = 90° + \theta_2 = 134.4°$

$\Sigma E_y = |\vec{E}_1| \cos(\beta_1) + |\vec{E}_2| \cos(\beta_2)$

$\Sigma E_y = 7.7 \times 10^6 N/C + (-7.7 \times 10^6 N/C)$

$\Sigma E_y = 0 N/C$



$\cos\theta_2 = \frac{5cm}{7cm}$

$\theta_2 = \cos^{-1}\left(\frac{5cm}{7cm}\right)$

$\theta_2 = 44.4°$

$\Sigma \vec{E}_p = (-1.57 \times 10^7 N/C)\hat{x} + (0)\hat{y}$

$|\Sigma \vec{E}_p| = \sqrt{\Sigma E_x^2 + \Sigma E_y^2} = 1.57 \times 10^7 N/C$